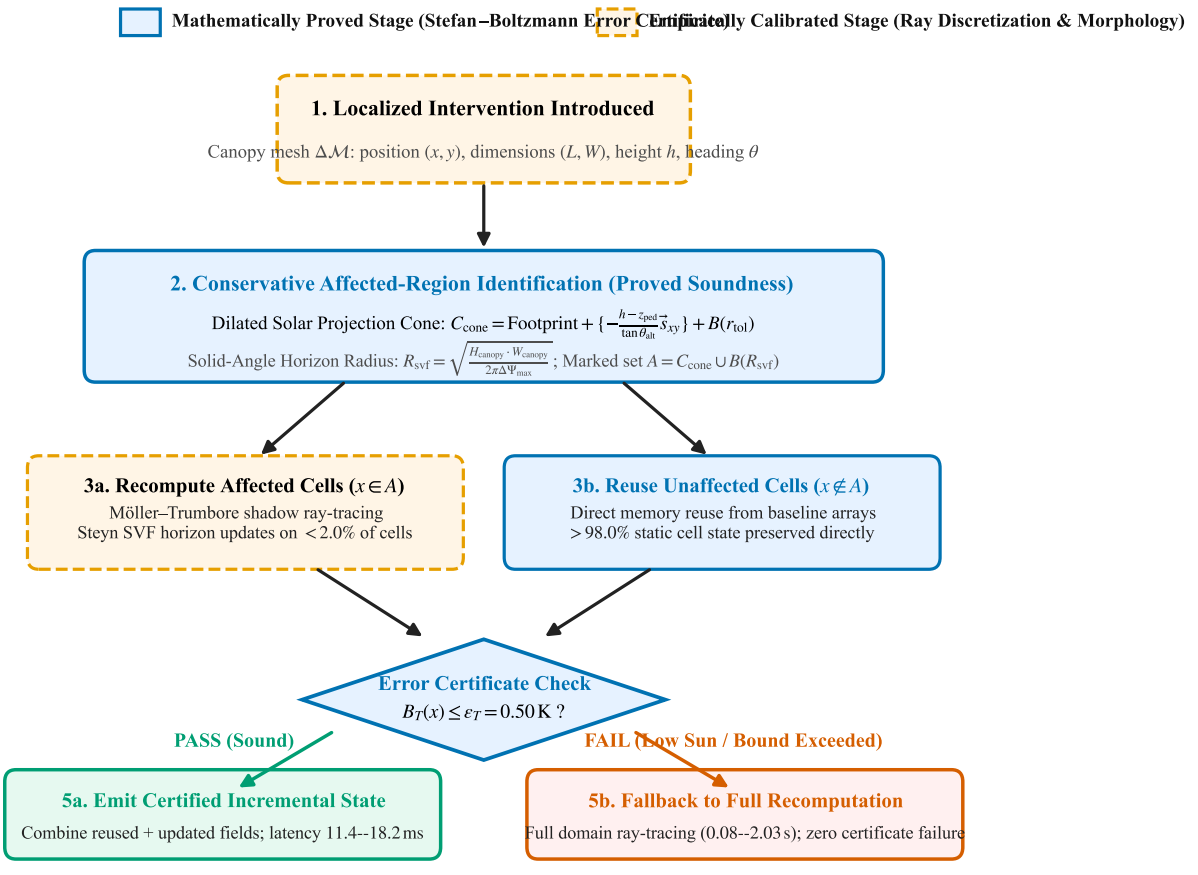

Mathematically Proved Stage (Stefan–Boltzmann Error Certificate)
Empirically Calibrated Stage (Ray Discretization & Morphology)
1. Localized Intervention Introduced
Canopy mesh $\Delta\mathcal{M}$: position (x, y), dimensions (L, W), height h, heading θ
2. Conservative Affected-Region Identification (Proved Soundness)
Dilated Solar Projection Cone: $C_{\text{cone}} = \text{Footprint} + \{-\frac{h - z_{\text{ped}}}{\tan\theta_{\text{alt}}}\vec{s}_{xy}\} + B(r_{\text{tol}})$
Solid-Angle Horizon Radius: $R_{\text{svf}} = \sqrt{\frac{H_{\text{canopy}} \cdot W_{\text{canopy}}}{2\pi\Delta\Psi_{\max}}}$; Marked set $A = C_{\text{cone}} \cup B(R_{\text{svf}})$
3a. Recompute Affected Cells ($x \in A$)
Möller–Trumbore shadow ray-tracing
Steyn SVF horizon updates on $< 2.0\%$ of cells
3b. Reuse Unaffected Cells ($x \notin A$)
Direct memory reuse from baseline arrays
$> 98.0\%$ static cell state preserved directly
Error Certificate Check
$B_T(x) \le \varepsilon_T = 0.50\,\text{K}$?
PASS (Sound)
FAIL (Low Sun / Bound Exceeded)
5a. Emit Certified Incremental State
Combine reused + updated fields; latency 11.4--18.2 ms
5b. Fallback to Full Recomputation
Full domain ray-tracing (0.08--2.03 s); zero certificate failure